

Smart MCQ Solver Challenge – Project Report

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1. Abstract

This report covers three models built for the Smart MCQ Solver Challenge (Kaggle), which requires ranking the top three most likely correct answers out of five options (A–E) per question, scored with Mean Average Precision at 3 (MAP@3): DeBERTa-v3-large + LoRA (pretrained), a Siamese BiLSTM (built from scratch), and RoBERTa-large + LoRA (additional model of choice).

Validation MAP@3 was near-ceiling for both transformers (DeBERTa 0.9983, RoBERTa 0.9983) and strong for the BiLSTM (0.9583 single-split, 0.9633 out-of-fold across 5 folds with augmentation + TTA). However, actual Kaggle public leaderboard scores (0.720–0.761 across all four submissions) sit far below every validation estimate, and the augmented 5-fold BiLSTM scored lower on the leaderboard than its simpler single-split version despite a higher validation score. Both point to train/validation leakage, most likely from 242 duplicate prompts and 183 fully-duplicate rows in the 2,000-row training set, as the dominant issue to resolve (Sections 3.1, 5). Note: the public leaderboard reflects only ~80% of the test data; final standings use the remaining 20% and may shift once the competition closes. Full details, code changes, and a reserved slot for wandb screenshots are below.

2. Introduction

The competition provides train.csv (2,000 rows: id, prompt, options A–E, answer) and test.csv (500 rows, unlabeled, same format). For every test question, a model must output up to three predicted labels in ranked order. MAP@3 scores a first-place correct answer 1.0, second-place 0.5, third-place 1/3, and an unranked correct answer 0.

Per the report guidelines, at least three distinct models are required: one pretrained, one built from scratch, and one additional model of choice. This report treats DeBERTa-v3-large + LoRA as the pretrained model (Section 4.1), the Siamese BiLSTM as the from-scratch model (Section 4.2), and RoBERTa-large + LoRA as the additional model of choice (Section 4.3), documented as a second pretrained backbone under an identical fine-tuning recipe, used to test whether Model 1's near-ceiling result is backbone-specific or a property of the dataset/approach.

3. Dataset, EDA & Tokenization

3.1 Dataset & EDA

train.csv has 2,000 rows and test.csv has 500, both with a prompt and five options (A–E). Diagnostics on train.csv surfaced three points worth flagging:

- Answer letters are mildly imbalanced: B 24.5%, C 23.0%, A 18.4%, D 17.9%, E 16.2%.
- 242 rows share a duplicate prompt with at least one other row, and 183 rows are fully duplicate (identical prompt and all five options).
- The correct option averages 181.9 characters, versus 162–167 characters average across each option column, a possible answer-length shortcut a model could exploit instead of genuine reasoning.

3.2 Preprocessing, Augmentation & Tokenization

The BiLSTM pipeline strips HTML/URLs and normalizes whitespace before tokenizing; the transformer pipelines pass prompt/option text directly to their pretrained tokenizers. The BiLSTM's improved run adds option-shuffle augmentation during training (options randomly reordered each epoch, label remapped, to

discourage positional shortcuts) and 5x test-time augmentation (TTA) at inference, averaging scores across 5 random option orderings.

DeBERTa-v3-large and RoBERTa-large both use their pretrained subword tokenizers (SentencePiece and byte-level BPE respectively); each question is encoded as five (prompt, option) sentence pairs (max length 256 tokens), batched via a custom DataCollatorForMultipleChoice that flattens the (batch, 5) structure for dynamic padding and reshapes it back before the model sees it. The BiLSTM instead uses a from-scratch regex tokenizer with a frequency-based vocabulary (minimum frequency 2, capped at 30,000, yielding 3,004 tokens from train+test text), since “built from scratch” rules out any pretrained tokenizer or embeddings.

4. Modeling & Experimentation

Models 1 and 3 share an identical LoRA fine-tuning recipe (only the backbone and target-module naming differ), applied via Hugging Face PEFT with AutoModelForMultipleChoice:

Hyperparameter	Model 1: DeBERTa-v3-large	Model 3: RoBERTa-large
LoRA rank / alpha / dropout	16 / 132 / 0.1	16 / 132 / 0.1
Target modules	query proj, value proj	query, value
Modules fully trained	classifier, pooler	classifier, pooler
Optimizer / LR	AdamW, 5e-5	AdamW, 5e-5
Batch size (effective)	2, accum 8 (eff. 16)	2, accum 8 (eff. 16)
Weight decay / warmup	0.05 / 10%	0.05 / 10%
Max sequence length	256 tokens	256 tokens
Epochs (submitted run)	8	20
Trainable parameters	2.62M / 437.7M (0.60%)	2.62M / 358.0M (0.73%)

4.1 Model 1: DeBERTa-v3-large + LoRA (Pretrained)

DeBERTa-v3-large (He, Gao & Chen, 2021) adds disentangled attention (separately encoding content and relative position) and an enhanced mask decoder to BERT-style architectures, pretrained with ELECTRA-style replaced-token detection. Used via AutoModelForMultipleChoice, which adds a scoring head producing one logit per (prompt, option) pair. The disentangled attention is well suited to MCQ answer selection, since it depends on the relative role of tokens rather than absolute position alone.

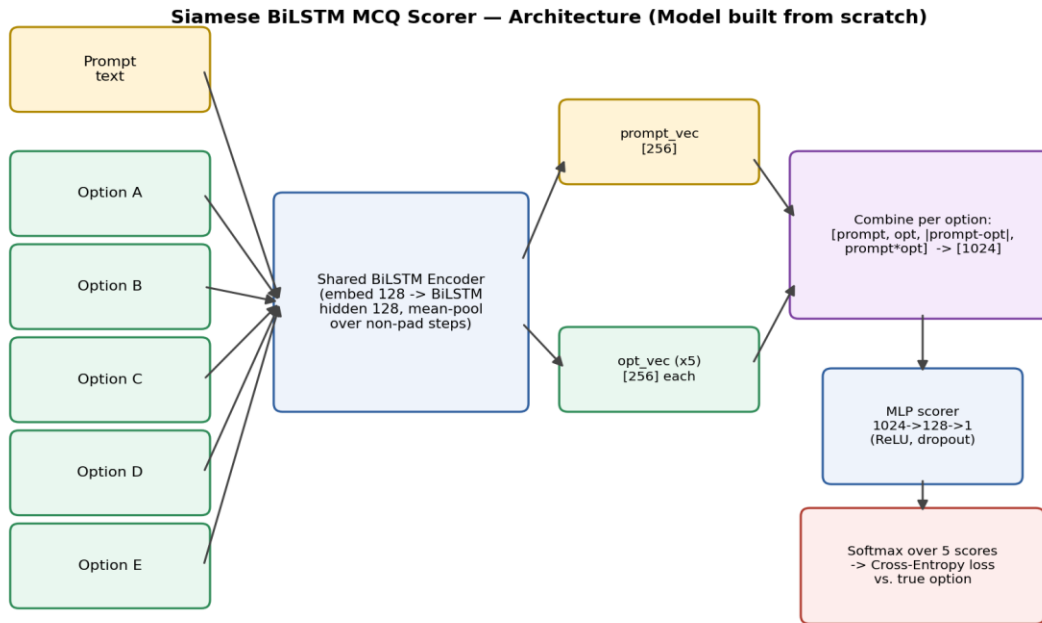
The submitted 8-epoch run, which matches the Kaggle score in Section 5, reached: eval_loss 0.147, accuracy 0.9967, MAP@3 0.9983, in approximately 26 minutes wall-clock.

Epoch	Train Loss	Val Loss	Val Accuracy	Val MAP@3
1	26.85	3.059	0.510	0.662
2	23.95	2.182	0.703	0.812
3	11.44	0.739	0.897	0.934
4	7.93	0.363	0.957	0.970
5	5.60	0.250	0.973	0.985
6	4.25	0.192	0.987	0.993
7	3.17	0.147	0.997	0.998

Epoch	Train Loss	Val Loss	Val Accuracy	Val MAP@3
8 (best)	3.47	0.155	0.997	0.998

Follow-up run: the notebook was re-run at 7 epochs with wandb logging enabled (project mcq-science-exam, run fek67miv), reaching eval loss 0.204, accuracy 0.99, MAP@3 0.995 in ~22 min. Not yet submitted to Kaggle

4.2 Model 2: Siamese BiLSTM (Built From Scratch)



A shared bidirectional LSTM (embedding 128, hidden 128, 1 layer, mean-pooled over non-pad steps) encodes the prompt and each option using the same weights. Per option, the prompt and option vectors combine into [prompt, option, |prompt – option|, prompt × option] (1,024-dim), reduced by an MLP scorer (1024→128→1, ReLU, dropout) to a single compatibility score; the five scores are softmaxed and trained with cross-entropy loss. No component uses pretrained weights (~0.78M total parameters, roughly 1/560th of the LoRA-adapted DeBERTa).

Trained two ways to test whether cross-validation and augmentation close the gap to the transformers:

- Run A (single split): 90/10 stratified split, no augmentation, no TTA. Early-stopped at epoch 8 (patience 4). Best val MAP@3 = 0.9583, macro-F1 = 0.9316.
- Run B (5-fold, augmented): 5-fold stratified CV with bagging, option-shuffle augmentation during training, and 5x TTA at inference. Out-of-fold MAP@3 = 0.9633, macro-F1 = 0.9396.

The OOF score aggregates predictions each made on data unseen by that fold's model, making it the more principled of the two estimates in isolation, though Section 5 shows this ranking reverses on the actual leaderboard:

Fold	1	2	3	4	5	OOF
Best Val MAP@3	0.9642	0.9600	0.9658	0.9671	0.9596	0.9633

4.3 Model 3: RoBERTa-large + LoRA (Additional Model of Choice)

RoBERTa-large (Liu et al., 2019) is a BERT-style encoder pretrained with dynamic masking, no next-sentence-prediction objective, and byte-level BPE tokenization, on a larger and more diverse corpus than the

original BERT. Chosen deliberately as a controlled comparison to Model 1, identical LoRA recipe, data pipeline, and tokenization strategy, changing only the backbone, to test whether Model 1's near-ceiling score reflects DeBERTa-v3's architecture specifically or is achievable by any sufficiently strong pretrained transformer under this recipe.

The submitted 20-epoch run reached: eval_loss 0.121, accuracy 0.9967, MAP@3 0.9983, matching DeBERTa to four decimal places on validation despite roughly 2.5x the epochs and ~80M fewer total parameters (358.0M vs. 437.7M). This near-identical validation outcome, reached by two architecturally different backbones through the same recipe, is itself informative: it suggests this dataset's validation ceiling under LoRA fine-tuning is more a property of the approach than of one specific architecture, though Section 5 shows this parity does not survive contact with the real leaderboard. Also now wandb-tracked under the same project (run 0ugx5029): wandb.ai/26ds2000018-iitmaana/mcq-science-exam/runs/0ugx5029

5. Performance & Kaggle Leaderboard

Primary metric: MAP@3, which rewards ranking the correct option earlier among the top 3 predictions. Accuracy (top-1 correct) and macro-F1 (BiLSTM only, from its confusion-matrix based error analysis) are supporting metrics reported alongside it. Per-epoch loss/MAP@3 curves come directly from each notebook's training logs (tables in Section 4); wandb dashboard screenshots for the newly-tracked runs can be added below.

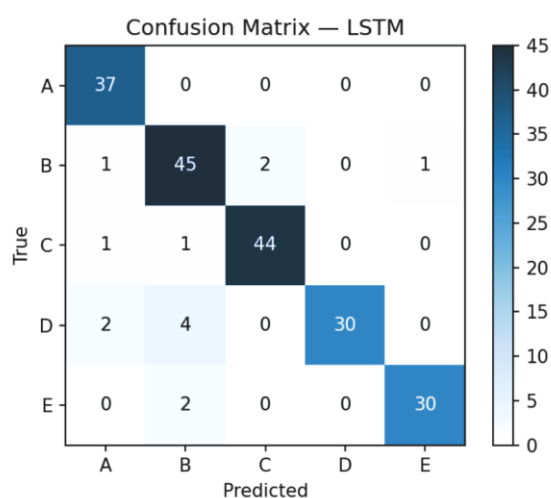
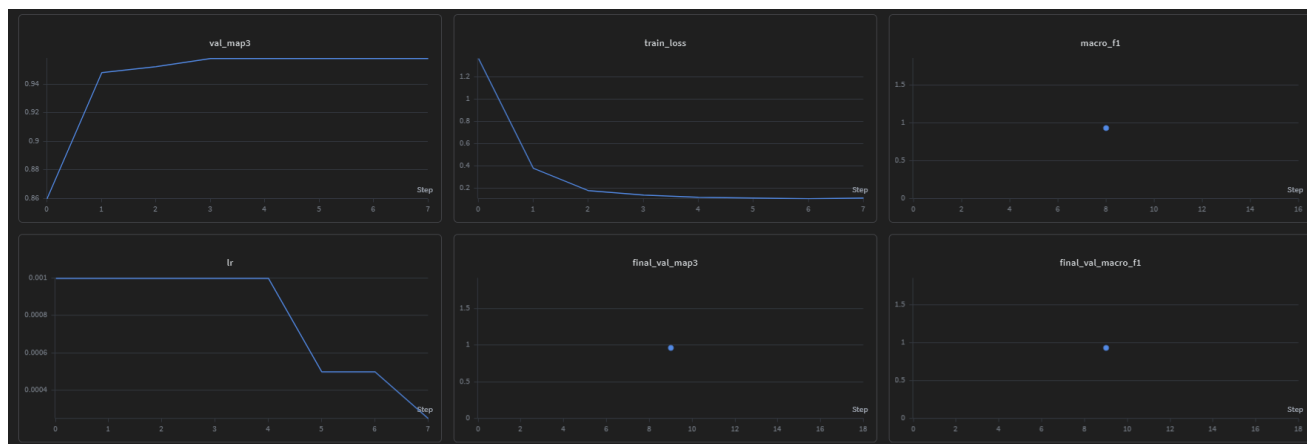
Deberta large v3 eval and train



Roberta-large



LSTM



Model / Submission	Val or OOF MAP@3	Kaggle Public MAP@3
DeBERTa-v3-large + LoRA (8-epoch, submitted)	0.9983	0.76101
RoBERTa-large + LoRA (20-epoch, submitted)	0.9983	0.74729
Siamese BiLSTM, single split	0.9583	0.72402
Siamese BiLSTM, 5-fold + aug + TTA	0.9633 (OOF)	0.71986

Note: this leaderboard is calculated on approximately 80% of the test data; final results use the remaining 20%, so standings above may still change once the competition closes.

Two patterns matter more than any single score above:

- Every validation/OOF score (0.958–0.998) sits far above every leaderboard score (0.720–0.761), roughly a 0.22–0.28 gap, too large for normal fine-tuning variance. The likely cause is the 242 duplicate prompts / 183 fully-duplicate rows (Section 3.1): the current split is label-stratified but not group-aware, so duplicates can land on both sides, letting models memorize rather than generalize.
- The 5-fold, augmented BiLSTM scored lower on the leaderboard (0.71986) than the simpler single-split version (0.72402) despite a higher OOF score (0.9633 vs. 0.9583), the opposite of what more rigorous cross-validation should produce. This suggests either the OOF estimate is still leakage-inflated, or the option-shuffle augmentation and/or TTA smoothed over a signal the real test set rewards.

- DeBERTa beat RoBERTa on the leaderboard by 1.4 points (0.761 vs. 0.747) despite an identical validation score (0.9983 each), further evidence that validation agreement between models does not guarantee equal generalization.

6. Conclusion & Future Work

6.1 Key Learnings

- LoRA adapts a large pretrained transformer to a new task while training under 1% of its parameters, reaching near-ceiling validation performance in a handful of epochs.
- Two different pretrained backbones (DeBERTa, RoBERTa), fine-tuned identically, converged to the same validation MAP@3, but this did not hold on the Kaggle leaderboard, where DeBERTa generalized measurably better.
- A from-scratch model (the BiLSTM, $\sim 1/560$ th the parameters of the transformers) can close much of the validation gap to a pretrained transformer through training procedure alone, cross-validation, augmentation, TTA, without any architecture change.
- The single biggest lesson: validation MAP@3 and Kaggle MAP@3 diverged by 0.22–0.28 points across all four models, a reminder that a validation split sharing rows with training data (via duplicates) can look excellent while revealing almost nothing about real generalization.

6.2 Challenges Faced

- 242 duplicate prompts / 183 fully-duplicate rows risk leakage in a label-stratified (not group-aware) split; the leaderboard-vs-validation gap is consistent with this leakage actually occurring, not just a theoretical risk.
- An apparent answer-length shortcut (correct options ~ 10 – 12% longer than distractors) may inflate all three models' scores independent of genuine reasoning.
- DeBERTa and RoBERTa were evaluated on a single train/validation split, unlike the BiLSTM's 5-fold OOF evaluation, an asymmetry in evaluation rigor across models that limits a fully fair comparison.

6.3 Future Work

- Highest priority: rebuild the split (and the BiLSTM's k-folds) to be group-aware, deduplicated by prompt, rather than only label-stratified, and re-measure all four models; this is likely to change which model looks best more than any architecture change would.
- Investigate the BiLSTM's fold reversal by ablating option-shuffle augmentation and TTA independently against the leaderboard, not just validation.
- Run a length-controlled ablation (e.g. truncating/padding options to equal length) to quantify how much of any model's MAP@3 comes from the length shortcut versus genuine reasoning.
- Build a weighted or rank-fusion ensemble of DeBERTa, RoBERTa, and the BiLSTM, best attempted after the leakage fix above, since ensembling three leakage-inflated models risks compounding rather than fixing the problem; pull the new wandb dashboards (Section 5) into this report; and consider quantizing DeBERTa/RoBERTa for cheaper inference once the leakage issue is resolved.

7. References

- He, P., Gao, J., & Chen, W. (2021). DeBERTaV3: Improving DeBERTa using ELECTRA-Style Pre-Training with Gradient-Disentangled Embedding Sharing. arXiv:2111.09543.
- Hu, E. J., et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685.
- Liu, Y., et al. (2019). RoBERTa: A Robustly Optimized BERT Pretraining Approach. arXiv:1907.11692.

Hugging Face model cards: microsoft/deberta-v3-large; roberta-large. Mangrulkar, S., et al., PEFT (Hugging Face). PyTorch docs: torch.nn.LSTM. Kaggle: Smart MCQ Solver Challenge. Weights & Biases documentation.